

Tutorial 3

Systems of linear equations, logic and elementary matrices

1. For each of the following statements say whether it is true or false, and give a reason for your answer.
 - (a) If $m = n$, then for every $A \in M_{m \times n}(\mathbb{R})$ and every m -vector $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution.
 - (b) If $m > n$, then for every $A \in M_{m \times n}(\mathbb{R})$ and every m -vector $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has at least one solution.
 - (c) If $m > n$, then for every $A \in M_{m \times n}(\mathbb{R})$ and every m -vector $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has no solution.
 - (d) If $m < n$, then for every $A \in M_{m \times n}(\mathbb{R})$ and every m -vector $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions.
2. Let U be a row echelon form of the augmented matrix of the system $A\mathbf{x} = \mathbf{b}$, where $A \in M_{m \times n}(\mathbb{R})$. Determine whether the following statements are true or false, and provide a reason for your answer.
 - (a) If U contains a row of 0's, the system $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions.
 - (b) If there are $n + 1$ pivots in U , the system $A\mathbf{x} = \mathbf{b}$ has a unique solution.
 - (c) If U does not contain a row of 0's, it is not in row echelon form.
3. Consider the following system of linear equations:

$$2x_2 + 4x_3 = 18$$

$$-x_1 - x_2 + 2x_3 = -4$$

$$2x_1 + 18x_3 = 16$$

$$2x_1 - 4x_2 + 10x_3 = -20$$

- (a) Write down the augmented matrix of this system. Is this matrix in row echelon form? Give a reason for your answer.
- (b) Use **the algorithm for Gaussian elimination that we've given you in the notes** to reduce the augmented matrix to row echelon form. Follow the steps in the algorithm **as given in the notes carefully, without taking short-cuts**. Stop when the algorithm tells you to do so. This should go *quickly*! (As you can see, it is important to consult your notes when doing the tutorial...)

- (c) Does this system have a solution? If so, write down the solution set of the system.
- (d) How would you now check your solution is correct? Do it!

In the next two questions you have to use some logic. Remember that if you believe a statement is *true*, you have to give a *proof* to show it is true; if you believe it is *false*, a *counter-example* is all that is needed to *disprove* it.

4. The statement

For every x , $P(x) \Rightarrow Q(x)$

has as its **converse** the statement

For every x , $Q(x) \Rightarrow P(x)$.

A statement and its converse are **NOT** logically equivalent. They mean quite different things.

Now consider the statement:

Every upper triangular matrix is a square matrix in row echelon form.

- (a) Is this statement true or false? Prove it or disprove it.
- (b) Write down the *converse* of this statement. Is the converse true or false? Prove it or disprove it.

5. The statement

For every x , $P(x) \Rightarrow Q(x)$

is logically equivalent to the statement

For every x , $\neg Q(x) \Rightarrow \neg P(x)$. (The symbol $\neg P$ means “not P” or “it is not true that P ”.)

The second statement is called the **contrapositive** of the first. A statement and its contrapositive **are** logically equivalent. They have exactly the same meaning.

Consider the statement:

For all real numbers k and all matrices A , if $kA = \mathbf{0}$ then $k = 0$ or $A = \mathbf{0}$.

Write down and then prove its contrapositive.

6. Which of the following matrices are elementary? Give reasons for your answers.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & -7 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

7. For each of the following two matrices, use **the definitions** to decide whether it is singular or non-singular:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

8. Consider the statement:

For every $n \times n$ matrix A , if $A^2 = \mathbf{0}$, then $A + I_n$ is invertible. (*)

- (a) Write down the converse of the statement (*).
- (b) Write down the contrapositive of the statement (*).
- (c) Prove or disprove the statement (*).
- (d) Prove or disprove the converse of the statement (*).